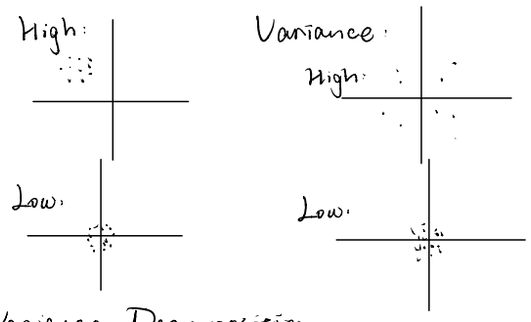
Bias and Variance

1. Generalization Error

Bias :



Bias-Variance Decomposition

Regression:

$D: \{x_i, y_i\}_{i=1}^{N}$

Determine the weight w.

$\Rightarrow \quad y = w^T \phi(x)$

New Statement:

$S = \{(x_i, y_i)\}_{i=1}^{m}$, where $y_i \in \mathbb{R}$

A hypothesis class $\mathcal{H}$ of regression functions.

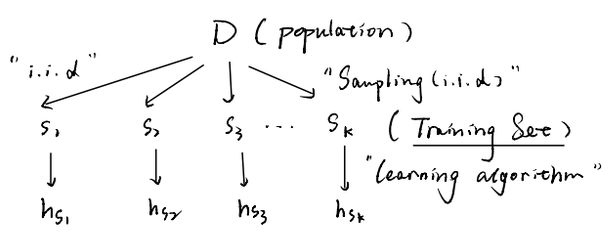
e.g. $\mathcal{H} = \{ h(x) = w^T \phi(x) \mid w \}$. "假设空间"

"w is arbitrary"

$\Rightarrow$ Choose one hypothesis h from $\mathcal{H}$.

Training Error

$$\hat{\epsilon}(h) = \frac{1}{m} \sum_{i=1}^{m} (y_i - h(x_i))^2$$



Training:

Objective: $\hat{h} = \arg\min_{h \in \mathcal{H}} \hat{\epsilon}(h)$

$\Rightarrow$ training error: $\hat{\epsilon}(\hat{h})$

Generalization Error:

$$\epsilon(h_S) = E_{(x,y)\sim D} [(y - h(x))^2]$$

Expectation (sampling (x,y) from D)

Generalization Gap.

$$\epsilon(h_S) - \epsilon(\hat{h}_S)$$

$$\epsilon = E_S [\epsilon(h_S)] = E_S [E_{(x,y)\sim D} [(y - h(x))^2]]$$

ϵ: Expected generalization error $\rightarrow$ We want to minimize it!

☆ The Bias-Variance Decomposition

$$\epsilon = E_S [E_{(x,y)} [(y - h_S(x))^2]]$$

1. $h_S(x) \rightarrow h_S$ for readability

$$\epsilon = E_S [E_{(x,y)\sim D} [(y - h_S)^2]]$$

$$= E_S E_{(x,y)} [(y - E_S(h_S) + E_S(h_S) - h_S)^2]$$

$$= E_S E_{(x,y)} [(y - E_S(h_S))^2] + E_S E_{(x,y)} [(E_S(h_S) - h_S)^2]$$

$$+ 2 E_S E_{(x,y)} [\underbrace{(y - E_S(h_S))}_{\text{scalar (for } E_S)} (E_S(h_S) - h_S)]$$

$$= E_S E_{(x,y)} [(y - E_S(h_S))^2] + E_S E_{(x,y)} [(E_S(h_S) - h_S)^2]$$

$$\boxed{+ 2 E_{(x,y)} [(y - E_S(h_S)) \cdot (E_S(E_S(h_S)) - E_S h_S)]}$$

☆ $E_S(E_S(h_S)) = E_S(h_S)$!

$\Rightarrow E_S(E_S(h_S)) - E_S h_S = 0$!

$$= E_S E_{(x,y)} [(y - E_S(h_S))^2] + E_S E_{(x,y)} [(E_S(h_S) - h_S)^2]$$

$$= \boxed{E_{(x,y)} [(y - E_S(h_S))^2]} + \boxed{E_x E_S [(E_S(h_S) - h_S)^2]}$$

"$(\text{Bias})^2$" "偏离程度" $\longleftrightarrow$ "Trade-off" $\longleftrightarrow$ "Variance" 离散程度

"Expected Generalization Error = Bias^2 + Variance!"

illustration: